

TEMPLATE

KEY PROJECT INFORMATION & VPA DESIGN DOCUMENT (PDD)

PUBLICATION DATE **04.05.2022**

VERSION **v. 2.0**

RELATED SUPPORT - [Programme of Activity requirements](#)

This document contains the following Sections

Key Project Information

Section A – Description of project

Section B – Application of approved Gold Standard Methodology (ies) and/or demonstration of SDG Contributions

Section C – Duration and crediting period

Section D – Summary of Safeguarding Principles and Gender Sensitive

Section E – Summary of Local stakeholder consultation

Section F – Eligibility and inclusion criteria for VPAs inclusion

Appendix 1 – Safeguarding Principles Assessment (mandatory)

Appendix 2 – Contact information of VPA Implementer
(mandatory)

Appendix 3 – Summary of Approved Design Changes (VPA specific)

KEY PROJECT INFORMATION

Type of VPA	☒ Real case VPA ☐ Regular VPA
Scale of VPA Note that a VPA can be of one scale. Please select applicable scale accordingly.	☒ Microscale ☐ Small scale ☐ Large scale
Title of corresponding real case VPA (if applicable)	GS1247 VPA 311 Guatemala Improved Cookstoves Project (GS11212) GS1247 VPA 312 Guatemala Improved Cookstoves Project (GS11213)
GS ID of real case VPA (if applicable)	GS11212 GS11213
GS ID of VPA	GS11212 GS11213
Title of VPA	GS1247 – Improved Kitchen Regimes Multi-Country -Guatemala Improved Cookstoves Project - VPA 311 GS1247 – Improved Kitchen Regimes Multi-Country - Guatemala Improved Cookstoves Project - VPA 312
Time of First Submission Date	14/07/2022
Date of Design Certification	N/A
Version number of the VPA-DD	54
Completion date of version	261 /10/2022
Coordinating/managing entity	Matthew Pike- CO2balance UK Ltd Maria Jimenez- CO2balance UK Ltd
VPA Implementer (s)	CO2balance UK Ltd
Project Participants and any communities involved	CO2balance UK Ltd
Host Country (ies)	Guatemala
GS ID and Title of applicable Design Certified VPA	N/A

GS ID and Title of applicable Performance Certified VPA Activity Requirements applied	N/A ☒ Community Services Activities ☐ Renewable Energy Activities ☐ Land Use and Forestry Activities/Risks & Capacities ☐ N/A
Other Requirements applied	N/A
Methodology (ies) applied and version number	GS MS Simplified Methodology for Efficient Cookstoves v1.1
Product Requirements applied	☒ GHG Emissions Reduction & Sequestration ☐ Renewable Energy Label ☐ N/A
VPA Cycle:	☐ Regular ☒ Retroactive

Table 1 – Estimated Sustainable Development Contributions

Sustainable Development Goals Targeted	SDG Impact (defined in Error! Reference source not found.)	Estimated Annual Average	Units or Products
13 Climate Action (mandatory)	Emissions Reductions	GS11212: 8531796 tCO2e GS11213 87809055 tCO2e	Emissions Reductions
3 Good Health and Well-Being	Total reduction in household air pollution for project activity in year y (%)	GS11212 62.0496% GS11213 56.4204%	HARPy
7 Affordable and Clean Energy	Number of project ICS active in year y ¹	GS11212 : 1,845 GS11213: 2,070	ICSa,y
1 No Poverty	Time saved collecting fuelwood	GS11212: 1.4 GS11213: 1.4	Hours/trip

¹ This ICSa,y accounts for Up,y in the Np,y (total stoves disseminated) figure. Np,y is set at 2,100. This is consistent against further SDG tools and SDG Impact spreadsheets provided.

SECTION A. DESCRIPTION OF PROJECT

A.1. Purpose and general description of project

CO2balance UK Ltd (CO2balance) and Estufas Chispa will distribute locally designed and manufactured stoves throughout Guatemala.

The regions of distribution are characterized by low penetration of clean cooking solutions and high wood fuel consumption as a result of current cooking methods. Nationally wood fuel accounts for 54% of fuel type for cooking².

The purpose of the Project is to reduce greenhouse gas emissions at a household level through the distribution of improved cookstoves which reduce wood consumption. As well as reducing greenhouse gas emissions, the Project will provide various benefits to the end user households and communities whilst helping to meet the UN Sustainable Development Goals (SDG) targets. Benefits include improved health through reduction in indoor air pollution, saving women time on unpaid chores, and reducing the income expenditure on fuelwood for stove recipients. The SDG impacts of the distribution of efficient cook stoves will be explored at greater detail, in later sections.

A.1.1. Eligibility of the VPA under approved PoA

This Project is automatically eligible for Gold Standard Certification as it follows the approved Gold Standard Activity Requirements 1.2(b) End-Use Energy Efficiency: efficient cooking, and eligible under the PoA as it is in one of the defined host countries (section A.3), Guatemala, and implements one of the defined project technologies (section A.4), improved cookstoves.

² INE, XII Censo Nacional de Población y VII de Vivienda – 2018- National Census on Population and Household

Table 2 Eligibility for VPA inclusion as per PoA requirements

No.	Eligibility Criterion	Description/ Required condition	Description of the VPA in relation to the criteria, Means of Verification and Supporting evidence for inclusion
1	Types of Project	Eligible Projects shall include physical action/implementation on the ground. Pre-identified eligible Project types are identified in the Eligibility Principles and Requirements section.	The projects involve the sale of locally designed and manufactured improved cookstoves. The project type is eligible under Community Services Activity Requirements 3.1.1(b) End-Use Energy Efficiency: efficient cooking.
2	Location of Project	Projects may be located in any part of the world.	The VPA is located in Guatemala as specified in Section A.2 of this document. Guatemala is one of the countries outlined in the PoA-DD. The VPA is defined by a range of GPS coordinates and current administrative maps to define the project boundary.
3	Project Area, Project Boundary and Scale	The Project Area and Project Boundary shall be defined. Projects may be developed at any scale although certain rules, requirements and limitations may apply under specific Activity Requirements, Impact Quantification Methodologies and Products Requirements. In order to avoid double counting the Project shall not be included in any other voluntary or compliance standards programme unless approved by Gold	This is a microscale project. The project boundary, including GPS co-ordinates and maps are included in Section A.2 and B.3. In order to avoid double counting, this VPA is not included by any other carbon standard. This project displaces baseline technologies, and at time of receipt individuals would not have access to any other emission reducing technologies. Our implementing partner isn't working with another

Standard (for example project, therefore counting through dual certification). the emission reductions (for example use of similar only once. technology or practices through which the potential arises for double counting or misestimation of impacts amongst projects). Additionally, each stove has a unique serial ID number which will help to identify it and avoid duplications.

The host country, Guatemala, and location of each VPA is specified in Section A.2, in line with the locations outlined in Section A.3 of the PoA-DD. The project boundary, including GPS co-ordinates and maps are included in Sections A.2 and B.3.

...	(d) Host Country Requirements	Projects shall be in compliance with applicable Host Country's legal, environmental, ecological and social regulations.	The Project Activity does not violate any of the Host Country's legal, environmental, ecological and social regulations.
	(e) Contact Details	As part of the Project Documentation the Project Developer shall provide (i) name and (ii) contact details of all Project Participants; AND in case of an organisation (iii) the legal registration details and (iv) documentation by the governing jurisdiction that proves that the entity is in good standing (defined as being a legal or other appropriate entity registered in or allowed to operate within the required jurisdiction and with no evidence of insolvency or legal/criminal notices placed against it or any of its Directors). Gold Standard retains the right (at its own discretion) to refuse use of the Standard where reputational concerns are highlighted.	Contact details of the Project Developer are included in Appendix 1.

(f) Legal Ownership	Full and uncontested legal ownership of any Products that are generated under Gold Standard Certification, (for example carbon credits) shall be demonstrated. Where such ownership is transferred from project beneficiaries this must be demonstrated transparently and with full, prior and informed consent (FPIC). Note that for certain Project types there is a requirement for full and uncontested legal land title/tenure to be demonstrated. These are contained within specific Activity or Product Requirements. All projects shall immediately report to Gold Standard any land title/tenure disputes arising.	At the point of sale, purchasers of the stove are informed of the transfer of carbon rights. The terms of the carbon transfer are stated on the CTF.
(g) Other Rights	As well as legal title and ownership, the Project Developer shall also demonstrate where required uncontested legal rights and/or permissions concerning changes in use of other resources required to service the Project (for example, access rights, water rights etc.). Any known disputes or contested rights must be declared immediately to Gold Standard by the Project Developer and resolved prior to further Project implementation in affected areas.	There are no disputes or contested rights that have been identified in relation to rights relevant to the project activity
(h) Official Development Assistance (ODA) Declaration	All Project Developers applying for project activities located in a country named by the	A declaration confirming that there is no diversion of ODA has been submitted.

OECD Development Assistance Committee's ODA recipient list and seeking Gold Standard Certification for carbon credits shall declare the Official Development Assistance (ODA) support. The Project Developer shall follow the GHG Emissions Reduction & Sequestration Product Requirements and submit the declaration at the time of Design Certification.

A.1.2. Legal ownership of products generated by the VPA and legal rights to alter use of resources required to service the project

In this project, CO2balance is the managing entity and is responsible for communication with the Gold Standard. CO2balance's implementing partner is Estufas Chispa who manufacture the stoves and market them. Estufas Chispa conduct the monitoring. CO2balance own the legal rights to the carbon credits created by the project.

The purchasers of the project stove are informed of the carbon transfer at the point of sale and receive a receipt which confirms the transfer of carbon rights. This entitles CO2balance to the rights of any carbon saved by the project.

A.2. Location of VPA

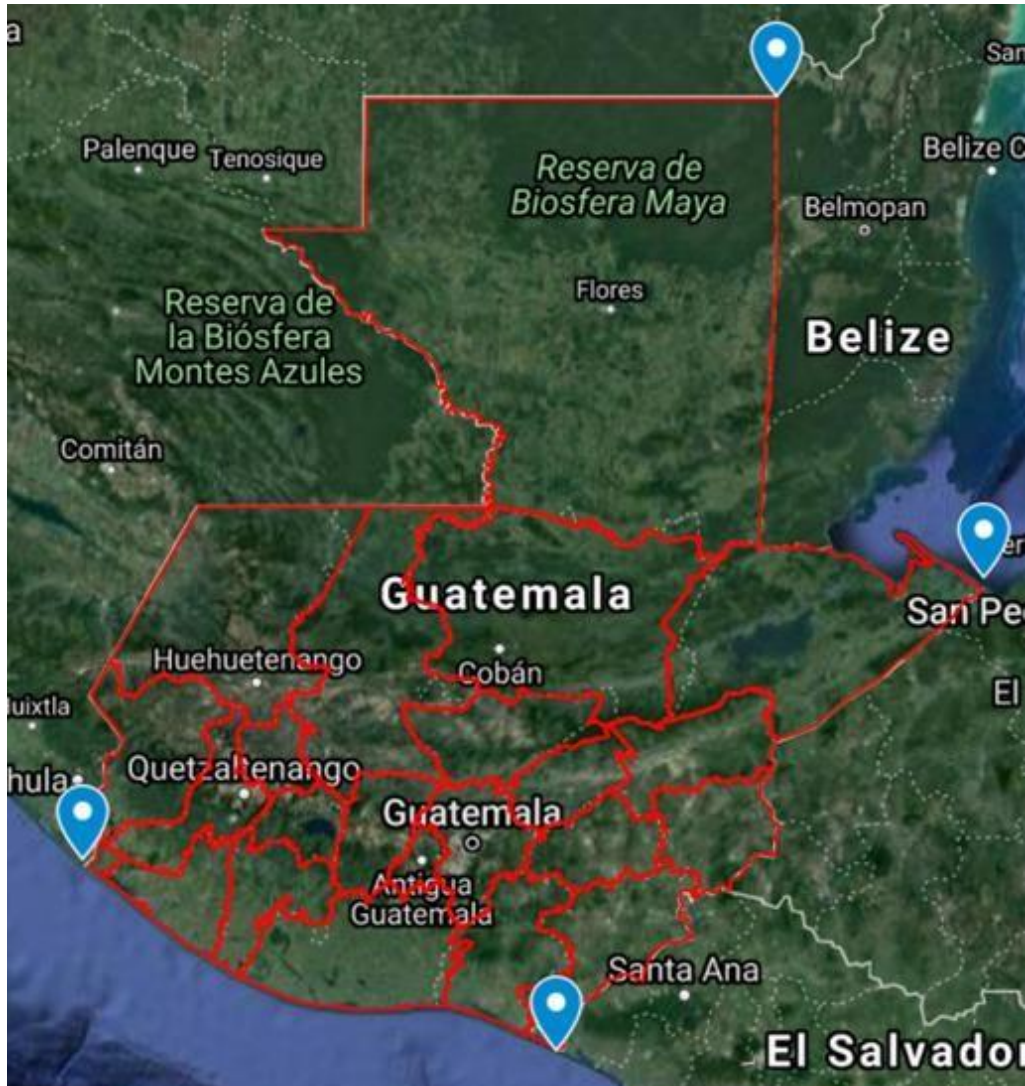


Figure 1: Republic of Guatemala - pins representing its northern, eastern, southern and western points.

Stove distribution is country-wide (Figure 1), however the country benefits from broad geographic and socioeconomic similarities throughout its districts. To demonstrate this and allow for representative baseline studies to cover the wider stove distribution locations, a stratification approach has been taken to group associated districts- this considered national census data on wood fuel use for cooking and forest cover. Resulting in 4 main groupings. High-Low rankings were based on below and above the averages. Baselines conducted in one department of the same strata group would be considered representative of all other departments of the same group.

Department	Forest Cover	Wood Use	Strata +/- Mean
Alta Verapaz	High	High	GROUP A
Baja Verapaz	High	High	GROUP A
Chimaltenango	High	High	GROUP A
Huehuetenango	High	High	GROUP A
Petén	High	High	GROUP A
Quiché	High	High	GROUP A
San Marcos	High	High	GROUP A
Sololá	High	High	GROUP A
Izabal	High	Low	GROUP B
Sacatepéquez	High	Low	GROUP B
Chiquimula	Low	High	GROUP C
Jalapa	Low	High	GROUP C
Retalhuleu	Low	High	GROUP C
Totonicapán	Low	High	GROUP C
El Progreso	Low	Low	GROUP D
Escuintla	Low	Low	GROUP D
Guatemala	Low	Low	GROUP D
Jutiapa	Low	Low	GROUP D
Quetzaltenango	Low	Low	GROUP D
Santa Rosa	Low	Low	GROUP D
Suchitepéquez	Low	Low	GROUP D
Zacapa	Low	Low	GROUP D

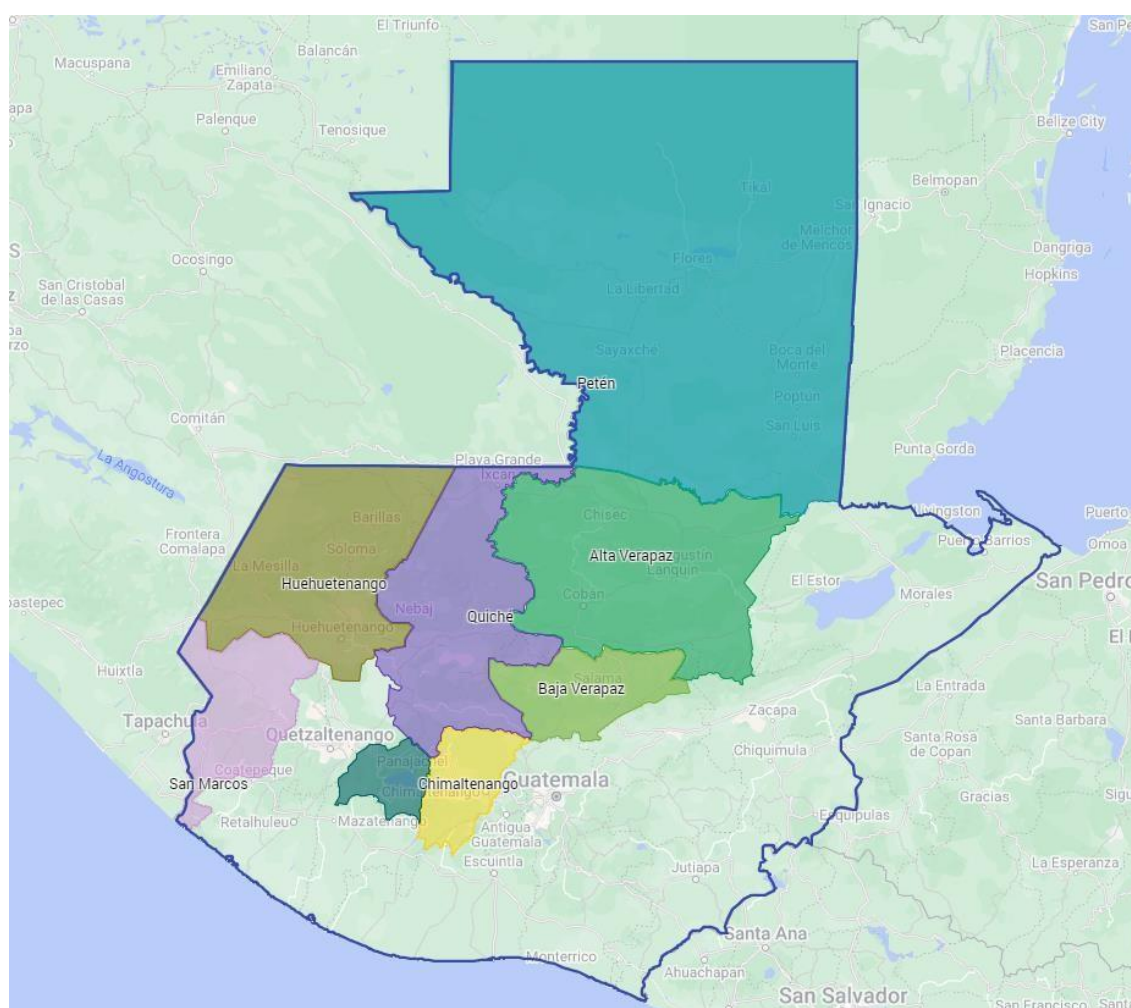


Figure 2: Image of Group A strata Departments within wider Guatemala border.

Table 3 Showing the latitude and longitude coordinates for each of the country's extremes.

	Latitude	Longitude
North	17.81585	-89.15813
East	15.75703	-88.23528
South	13.72785	-90.13592
West	14.54816	-92.24529

A.3. Technologies and/or measures

Households in Guatemala often cook on traditional three stone fires. These are extremely inefficient and demand a lot of firewood, resulting in high CO₂ emissions, deforestation, and exposure to indoor air pollution. The project stoves will displace the inefficient baseline technology and reduce CO₂ emissions by

reducing wood use.

The reduced use and demand for firewood will reduce greenhouse gas emissions, reduce demand of wood collection on local forests (for those that collect wood), reduce requirement to purchase wood (for those who buy wood), reduce the time spent collecting wood and cooking, and reduce exposure to harmful indoor air pollution.

The project stoves are designed and manufactured locally by Estufas Chispa.



Figure 3 Chispa Jumbo



Figure 4 Chispa Plancha

The stove uses wood and all parts are durable, but easily replaceable should breakdowns occur. The lifespan of the stove is from 3 to 5 years for the chimney, and 10 years for the body of the stove. The griddle is expected to last 5 years. If the stove breaks down earlier, users can contact our in-country partner through the continuous grievance mechanism.

All stoves of the project will have a guarantee of 3 years. Users can contact Estufas Chispa through the Customer Service WhatsApp line, to get a replacement if any part breaks. This contact number is easily accessible and found next to the stove ID of the stove.

Each stove shall have a unique serial number to distinguish it from other stoves, when it is sold the householder is given a receipt identify it and to avoid double counting. The serial number is equal to the recipient's personal national ID number. There are two stove types that will be distributed as part of the project.

The stove efficiency of the Chispa Plancha is 38.5% The stove efficiency of the Chispa Jumbo is 31.5%. These efficiencies were obtained as an average between the cold and hot start efficiency of each model. This definition is in line with the Gold Standard Water Boiling Test for Efficiency Determination. . The efficiencies were certified through lab test carried out by the University of San Carlos in Guatemala, and signed by their head Engineer.

As both technologies have different efficiencies, they will be monitored separately.

The report showed other key parameters being as follows:

	Chispa Plancha (stove 1)	Chispa Jumbo (stove 2)
Efficiency	38.5%	31.5%
CO level	98.14% reduction	97.34% reduction
PM 2.5	98.75% reduction	96.95% reduction
CO2 level	98.68% reduction	97.23% reduction
Power	3,876 Kw	4.191 Kw

This VPA will positively contribute to the following SDGs:

SDG 3 Good Health and Wellbeing: This VPA involves the distribution of improved cookstoves to households using inefficient three-stone fires as their primary cooking method. By switching to an improved cooking technology, this VPA reduces the amount of non-renewable biomass used for cooking, thus reducing household air pollution (HAP).

SDG 7 Affordable and Clean Energy: The project distributes improved cookstoves to replace the inefficient 3-stone fire. It increases the proportion of the population relying on clean cooking technology, especially in rural areas. The stoves are distributed to the local communities and subsidized to ensure greater access within the communities.

SDG 1 No Poverty: The project reduces the wood dependency, as the Project Stove needs less fuel than the baseline technology. Users will gain time that was previously spent collecting firewood, which they can invest in other activities that contribute to their wellbeing.

SDG 13 Climate Action: The projects will meet SDG 13 by realizing a real reduction in CO2e emissions.

A.4. Scale of the VPA

This is a micro-scale project. VERs are capped at 10,000 per year per VPA.

A.5. Funding sources of VPA

There is no public or ODA funding for this project activity, all revenue for the project will be derived from the sales of VERs.

SECTION B. APPLICATION OF APPROVED GOLD STANDARD METHODOLOGY (IES) AND/OR DEMONSTRATION OF SDG CONTRIBUTIONS

B.1. Reference of approved methodology (ies)

GS MS Simplified Methodology for Efficient Cookstoves v1.1.

B.2. Applicability of methodology (ies)

The project is expressly applicable under the Improved Cookstove Methodology.

Methodology Requirement	Project
1. This methodology is applicable, i. If the baseline fuel is only fire wood; and ii. If the baseline stove is a three stone fire, or a conventional device without a grate or a chimney i.e., with no improved combustion air supply or flue gas ventilation: and iii. If the project stove is single pot or multi pot portable or in-situ cook stoves with specified efficiency of at least 20%.	Baseline show a prevalence of three stone fire use, and rudimentary stoves that are that are fueled through firewood. Thermal efficiency values averaging 11% for three stone fires and 18% for rudimentary stoves% have been independently tested and verified by San Carlos National University lab. Therefore, the rudimentary stoves cannot be considered improved cookstoves because their efficiency does not reach 20%. For the 2.4% that use LPG, this is <i>de minimus</i> . This amount will be discounted from the Emission Reductions in each monitoring period.
2. The project boundary can be clearly identified, and the cookstoves counted in the proposed project activity are not included in another voluntary market or CDM project activity (i.e. no double counting takes place). The project proponent must have a mechanism in place together with appropriate mitigation measures to prevent double counting	Boundaries of project distribution area clearly marked and described in Section A2. No stoves distributed outside this project boundary will be included in the project.
3. The project proponent must clearly communicate that the entity is claiming ownership rights and selling of the emission reductions resulting from the	Transfer of carbon rights is stated in the CTF.

project activity. This must be communicated to the efficient cookstoves producers, retailers and end users by contract or clear written assertions in the transaction paperwork.	The LSC has been postponed due to Covid-19. A virtual LSC took place on 06/10/2022 explaining the transfer of carbon rights. Once an in person LSC is possible, this will be held and re-explained in person.
4. The use of the baseline technology as a backup or auxiliary technology in parallel with the improved technology introduced by the project activity is permitted as long as a mechanism is put into place to encourage the removal of the old technology and the definitive discontinuity of its use.	The project stove is purchased by users as they see some benefit in using it to save on fuel use as well as improved health benefits over their current methods, such as saving on wood costs, saving time on wood collection, ease of use, and/or reduced smoke. At time of distribution, users are technically trained in person on how to install, use and maintain their project stove. In this training, awareness on the negative impacts of three stone fires is raised, encouraging the removal of these baseline technologies. Parallel baseline technology use will be revealed during annual monitoring (Kitchen Survey and Usage survey) and be determined by surveys and hence used to account for parallel baseline and project technology use. Annual Monitoring results will be shared in Verification Reviews.

B.3. VPA boundary

VPA boundaries are explained in section A.2.

Source		GHGs Included?	Justification/Explanation
Baseline Scenario	Combustion of firewood	CO ₂ Yes	Important source of emissions
		CH ₄ Yes	Important source of emissions
		N ₂ O Yes	Can be significant for some fuels

Project scenario		CO ₂	Yes	Important source of emissions
	Combustion of firewood	CH ₄	Yes	Important source of emissions
		N ₂ O	Yes	Can be significant for some fuels

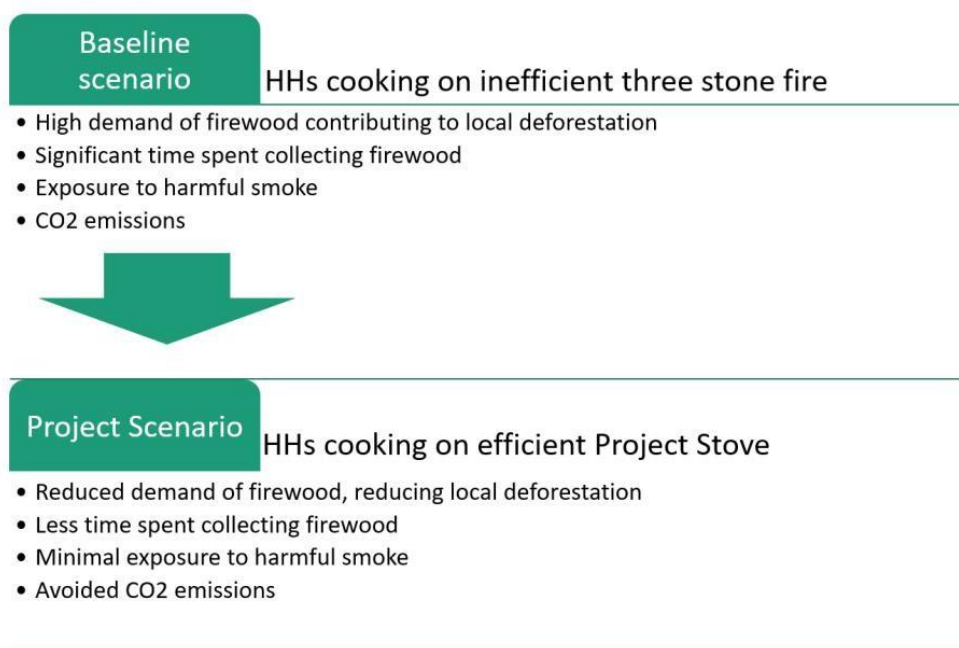


Figure 5: Project Activities Flow Chart.

B.4. Establishment and description of baseline scenario

The baseline study was conducted in two phases, one in March 2022 and one in June 2022. In March, 163 surveys were conducted, and in June 214 surveys were conducted. We chose to have a larger sample than required to make it more representative. The baseline surveys follow the requirements outlined in Improved Cookstove Methodology, following the stratification grouping approach described in Section A.2.

The baseline surveys were outsourced to experienced organisations.

All baseline surveys were conducted on households who will be receiving a project stove. Instead of randomly selecting, resulting in fewer subjects to survey, we have surveyed all households due to receive a stove in Union Cantinil, Alta Verapaz, Quiche and Solola.

The surveys contain 30 questions regarding demographics, household characteristics, kitchen habits, fuel use and health.

The results of the baselines show 77.96% use of three stone fire for cooking, and 18% use of a rudimentary stove. The rudimentary stove is in-built in household

kitchens, and shows an efficiency similar to a three stone fire. Baselines show that most of their chimneys are not functioning.

There is also a 2,4% use of LPG, which will be discounted from the Emission Reductions in every monitoring period as a self-imposed measure.

Find more information on the poor performance of rudimentary stoves here:

- <https://cleancooking.org/reports-and-tools/emission-and-efficiency-of-improved-woodburning-cookstoves-in-highland-guatemala/>
- <https://www.sciencedirect.com/science/article/pii/S0973082608602392>
- https://pdf.usaid.gov/pdf_docs/PDACM099.pdf



Figure 6 Rudimentary cookstove in-built in some houses.

The Kitchen Performance test was conducted on three stone fires in situ. 45 KPTs were completed over the course of 4 days. Results showed average wood usage of 15.5 kg/hh/day. Three stone fires are extremely inefficient and demand large quantities of fuelwood, resulting in high greenhouse gas emissions, deforestation and exposure to indoor air pollution, as shown by the KPTs. For this project, the conservative methodology default of 0.5 tonnes/capita/year has been chosen.

The baseline study also found the following information. A mix of collection and purchase of firewood is present. 55% of households only buy fuelwood, 17% collect and 28% do both.

The Baseline Kitchen Survey shows that households spend on average 2.8h collecting firewood per trip, and take 3 trips per week. As the project stove reduces firewood consumption, the burden of collecting firewood will be significantly reduced.

On average, families spend 4,23 hours per day cooking. This presents a high exposure to indoor air pollution caused by the smoke. The project stove reduces cooking time and smoke pollution. Therefore, the project will reduce cases of respiratory illnesses caused by the inhalation of smoke.

The project stoves will displace the inefficient baseline technology and reduce greenhouse gas emissions by 58.00%. The reduced use and demand for firewood will reduce greenhouse gas emissions, reduce demand of wood collection on local forests (for those that collect wood), reduce requirement to purchase wood (for those who buy wood), reduce the time spent collecting wood and cooking, and reduce exposure to harmful indoor air pollution.

B.5. **Demonstration of additionality**

Specify the methodology, activity requirement or product requirement that establishes deemed additionality for the proposed project (including the version number and the specific paragraph, if applicable).	As demonstrated in the Gold Standard for the Global Goals Community Services Activity Requirements section 4.1.9 – Projects that meet any of the following criteria are considered as deemed additional and therefore are not required to prove Financial Additionality at the time of Design Certification: (a) Positive list (Annex B) (b) Projects located in LDC, SIDS, LLDC © Micro-scale projects
Describe how the proposed project meets the criteria for deemed additionality.	The project meets criterion 3 as it is registered as a Micro-scale project.

The project is therefore additional.

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B.5.1. Prior Consideration

Discussions for this project began in early 2020 between CO2balance and Estufas Chispa. Factors outside the project control, for example the impacts of COVID, across aspects of the project has resulted in some delays to project registration and approach compared to what had originally been intended.

B.5.2. Ongoing Financial Need

N/A

B.6. Sustainable Development Goals (SDG) outcomes

Sustainable Development Goals Targeted	Most relevant SDG Target	SDG Impact Indicator (Proposed or SDG Indicator)
SDG 3. – Good Health and Well Being	3.9: By 2030, substantially reduce the number of deaths and illnesses from hazardous chemicals and air, water and soil pollution and contamination	HAPR _y : total reduction in household air pollution for project activity in year y (%)
SDG 1. No Poverty	1.2.2 Proportion of men, women and children of all ages living in poverty in all its dimensions according to national definitions	Try: Time Reduction in collecting firewood per trip in year y
SDG 7. – Affordable and Clean Energy	7.2: By 2030, increase substantially the share of renewable energy in the global energy mix	ICS _{a,y} : number of project ICS active in year y
SDG 13. - Climate Action (mandatory)	13.B.1: Number of least developed countries and small island developing States with nationally determined contributions, long-term	Emissions Reductions

strategies, national adaptation plans, strategies as reported in adaptation communications and national communications

B.6.1. Explanation of methodological choices/approaches for estimating the SDG Impact

SDG 1: No Poverty

The project stove requires less fuel, therefore making households less dependent on spending time collecting wood. The burden of collecting fuel is heavy, as it takes a significant time of their days, often several hours per trip

The average % decrease per household in time spent collecting fuel will be taken as a proxy contribution towards the SDG³. The overall percentage reduction in time spent collecting firewood by the project activity is then calculated as follows:

$$TR_y = (T_{b,y} - T_{p,y}) / T_{b,y}$$

Where:

Tr_y = Total reduction time spent collecting firewood for project activity in year y(%)

$T_{b,y}$ = Baseline time spent collecting firewood per household per trip (hours)

$T_{p,y}$: Project time spent collecting firewood per household per trip (hours)

It is predicted that time spent collecting firewood will be reduced as a result of the project⁴. To infer as to what project participants are doing with their time saved from the project, qualitative questions will be included in the monitoring surveys which ask respondents how they spend their time saved and answers will be divided into designated time use categories. In some circumstances, it may be the case where respondents comment on the tasks they undertook in their spare time and these are recorded by field staff.

SDG 3 (Good Health and Wellbeing)

³ <https://cleancooking.org/the-value-of-clean-cooking/>

⁴ Petrokofsky, G.; Harvey, W.J.; Petrokofsky, L.; Ochieng, C.A. 2021. 'The Importance of Time-Saving as a Factor in Transitioning from Woodfuel to Modern Cooking Energy Services: A Systematic Map.' *Forests*, 12, 1149. [Accessible at: <https://doi.org/10.3390/f12091149>].

Switching from an unimproved stove to an improved cookstove reduces the volume of non-renewable biomass that needs to be burned for the household's daily cooking needs. Therefore, the reduction in non-renewable biomass burned can be taken as a proxy indicator of how these VPAs contribute towards the SDG 3 target of substantially reducing the number of deaths and illnesses from hazardous chemicals and air, water and soil pollution and contamination.

The average difference in fuel use will be used to calculate the % decrease in fuel use, which is the figure used to calculate the VPAs' contribution to the SDG target.

The overall reduction in HAP by the project activity are then calculated as follows:

$$\text{HAPR}_y = ((\text{Bb},y - \text{Pp},y) / \text{Pb},y) * \text{Up},y$$

Where:

HAPR _y	Total reduction in Household Air Pollution for project activity in year y (%)
Bb,y	Quantity of fuel that is consumed in the baseline scenario b during year y (kg/household-day)
Pp,y	Quantity of fuel that is consumed in the project scenario p during year y (kg/household-day)
Up,y	Usage rate in project scenario p during year y

SDG 7 (Affordable and Clean Energy)

The VPA involves the distribution of ICS, a fuel-efficient technology, to replace the inefficient baseline cooking technologies.

To measure 7.1 (universal access to affordable, reliable and modern energy services), indicator 7.1.2 provides for the measurement of the 'proportion of the population with primary reliance on clean fuels and technology'. The outcome for SDG 7 is quantified as the additional number of persons having access to efficient cookstove technology in the project activity and is calculated using parameters from the project scenario.

$$\text{ICS}_{a,y} = \text{ICS}_{t,y} * \text{U}_{p,y}$$

Where:

ICS _{a,y}	Project ICS active in year y
ICS _{t,y}	Project ICS distributed in year y

$U_{p,y}$ Usage rate

SDG 13 (Climate Action)

The contribution to SDG 13 will be measured through carbon credits issued following the Gold Standard Simplified Cookstove 1.1 methodology.

$$ER_y = \sum_{0101}^{x10y} N_{p,y} * P_y * U_{p,y} * (f_{NRB,y} * EF_{b,fuel,CO_2} + EF_{b,fuel,non_CO_2}) * (1 - DF_{b,Stove,y})$$

Total emissions reductions are calculated as follows, outlined in the methodology:

Where:

NP,y = Number of project cookstoves of each age group operational in the year y

P_y= Quantity of fire wood that is saved in the year y (tonnes per household in year y)

UP,y= Usage rate for project cookstoves in year y, based on adoption rate and drop off rate revealed by usage surveys (fraction)

fNRB,b,y = Fraction of biomass, used in year y for baseline scenario, which can be established as non- renewable. The project proponents shall estimate project specific national/ regional value⁴ or apply the default fNRB value provided by the CDM Executive Board and endorsed by the host country DNA .

EF_{b,fuel,CO2} = CO2 emission factor of fire wood that is substituted or reduced. (Default value for wood fuel 1.747 tCO2/ton of wood).

EF_{b,fuel,non_CO2} = Non-CO2 emission factor of fire wood that is substituted or reduced. (Default value for wood fuel 0.53 tCO2/ton of wood) .

DF_{b,Stove,y}= Usage of baseline cookstove during the year y (fraction) in project scenario.

Y = Year of the crediting period

X = $y - 1$

Quantity of fire wood that is saved (P_y) is estimated as follows:

$$P_y = B_{b,y} * (1 - \eta_b / \eta_{p,y}) \dots \dots \dots (2)$$

Where:

$B_{b,y}$ = Quantity of fire wood consumed in baseline scenario during year y (tonnes per household per year)

$\eta_{p,y}$ = Efficiency of project cookstove in year y (fraction)

η_b = Efficiency of the baseline cookstove being replaced (fraction). A default value of 10% shall be used if the replaced cookstove is a three stone fire, or a conventional device without a grate or a chimney i.e., with no improved combustion air supply or flue gas ventilation

B.6.2. Data and parameters fixed ex ante

SDG 1

Data/parameter	$T_{b,y}$
Unit	Hours
Description	Baseline time spent collecting firewood per trip
Source of data	Baseline survey
Value(s) applied	2.8
Choice of data or Measurement methods and procedures	Deemed valid by Methodology.
Purpose of data	Calculation of time reduction, which can then be invested into other activities that contribute to their wellbeing

Additional comment	
	-

SDG13

Data/parameter	$EF_{b,fuel,CO_2}$
Unit	tCO ₂ /tonne of firewood
Description	CO ₂ emission factor arising from use of fire wood in baseline scenario (includes production, transport and use)
Source of data	1.747 tCO ₂ /ton of fire wood, IPCC default values, table 1.4 of chapter 1 of Vol. 2, 2006 IPCC Guidelines for National Greenhouse Gas Inventories
Value(s) applied	1.747
Choice of data or Measurement methods and procedures	Deemed valid by Methodology.
Purpose of data	Calculation of baseline emissions.
Additional comment	
	-

Data/parameter	$EF_{fuel,non-CO_2}$
Unit	tCO ₂ /tonne of fire wood
Description	Non-CO ₂ (CH ₄ and N ₂ O) emission factor arising from use of wood fuel in baseline scenario
Source of data	0.53 tCO ₂ /tonne of fire wood, IPCC default values, Table 2.9 of Chapter 2 of Vol. 2, 2006 IPCC Guidelines for National Greenhouse Gas Inventories
Value(s) applied	0.53

Choice of data or Measurement methods and procedures	Deemed valid by Methodology using the emission factors in the table below.
Purpose of data	Calculation of emission reductions.
Additional comment	-

Gas	Default Emissions factor (kg_gas/TJ _{NCV})	GWP of gas	Default Emissions factor (kg_CO ₂ e/TJ _{NCV})	Default Emissions factor (t_CO ₂ e/TJ _{NCV})
CH ₄	300	28	8,400	8.4000
N ₂ O	4	265	1,060	1.060
			Total	9.460

Data / Parameter	$f_{NRB,i,y}$
Unit	Fractional non-renewability
Description	Non-renewability status of woody biomass fuel in scenario i during year y
Source of data	Calculated in line with CDM Tool
Value(s) applied	0. 6973
Measurement methods and procedures	CDM Tool
Monitoring frequency	Fixed for the project unless reassessed in future as per PP requirements.
QA/QC procedures	
Purpose of data	Calculation of emission reductions
Additional comment	-

SDG 3:

	Bb,y
	tonnes
	Baseline indoor air pollution – average baseline yearly fuel use, per technology (tonnes)
	Baseline KPT

Value(s) applied	5.79
Choice of data or Measurement methods and procedures	
Purpose of data	To demonstrate the reduced incidence of indoor air pollution, a major cause of illness and death in Guatemala, through the reduction in wood fuel combustion.
Additional comment	To

No SDG 7 data and parameters fixed ex-ante.

B.6.3. Ex ante estimation of SDG Impact

SDG 1: GS11212 & GS11213

$$Try = Tby - Tpy$$

SDG 1: No Poverty	Description	$TRY = (Tb,y - Tp,y) / Tb,y$	Unit	Source
TRy	Total reduction time spent collecting firewood for project activity in year y	1.4	hours	
Tb,y	Baseline time spent collecting firewood per household per trip	2.8	hours	Baseline Survey
Tp,y	Project time spent collecting water and firewood per household per trip	1.4	hours	Estimation

SDG3: GS11212

$$HARPy = ((Pb,y - Pp,y) / Pb,y) * Up,y$$

SDG 3: Good Health and Well Being	Description	$HARPy = ((Pb,y - Pp,y) / Pb,y) * Up,y$	Unit	Source
HARPy	Total reduction in Household Air Pollution for project activity in year y (%)	62.06%	%	
Pb,y	Quantity of fuel that is consumed in the baseline scenario b during year y (kg/household-day)	5.82	tonnes/hh/year	bKPT
Py	Quantity of firewood that is saved in year y	4.02	tonnes/hh/year	Ex antes
Pp,y	Quantity of fuel that is consumed in the project scenario p during year y (kg/household-day)	1.81	tonnes/hh/year	
Up,y	Usage rate in project scenario p during year y	90%	%	

SDG 3: Good Health and Well Being	Description	$HARPy = ((Pb,y - Pp,y) / Pb,y) * Up,y$	Unit	Source
HARPy	Total reduction in Household Air Pollution for project activity in year y (%)	62.49%	%	
Bb,y	Quantity of fuel that is consumed in the baseline scenario b during year y (kg/household-day)	5.79	tonnes/hh/year	bKPT
Py	Quantity of firewood that is saved in year y	4.02	tonnes/hh/year	Ex antes
Pp,y	Quantity of fuel that is consumed in the project scenario p during year y (kg/household-day)	1.77	tonnes/hh/year	
Up,y	Usage rate in project scenario p during year y	90%	%	

GS11213

TEMPLATE- VPA Design Document

SDG 3: Good Health and Well Being	Description	HARPy = ((Pb,y - Pp,y)/Pb,y)*Up,y	Unit	Source
HARPy	Total reduction in Household Air Pollution for project	56.04%	%	
Bb,y	Quantity of fuel that is consumed in the baseline scenario	5.83	tonnes/hh/year	bKPT
Pp,y	Quantity of firewood that is saved in year y	3.63	tonnes/hh/year	Ex antes
Pp,y	Quantity of fuel that is consumed in the project scenario	2.20	tonnes/hh/year	
Up,y	Usage rate in project scenario p during year y	90%	%	

SDG 3: Good Health and Well Being	Description	HARPy = ((Pb,y - Pp,y)/Pb,y)*Up,y	Unit	Source
HARPy	Total reduction in Household Air Pollution for project	56.42%	%	
Bb,y	Quantity of fuel that is consumed in the baseline scenario	5.79	tonnes/hh/year	bKPT
Pp,y	Quantity of firewood that is saved in year y	3.63	tonnes/hh/year	Ex antes
Pp,y	Quantity of fuel that is consumed in the project scenario	2.16	tonnes/hh/year	
Up,y	Usage rate in project scenario p during year y	90%	%	

SDG 7:

GS11212

Active project stoves = $N_{p,y} * U_{p,y}$

SDG 7: Affordable and Clean Energy	Description	$N_{p,y} * U_{p,y}$	Unit	Source
Active ICS disseminated by project	The number of active ICS disseminated by the project	1845	Number	
$N_{p,y}$	Number of stoves distributed	2050	Number	Ex antes
$U_{p,y}$	Usage rate in project scenario during year y	90%	%	

GS11213

SDG 7: Affordable and Clean Energy	Description	$N_{p,y} * U_{p,y}$	Unit	Source
Active ICS disseminated by project	The number of active ICS disseminated by the project	2070	Number	
$N_{p,y}$	Number of stoves distributed	2300	Number	Ex antes
$U_{p,y}$	Usage rate in project scenario during year y	90%	%	

VPA stove numbers will be redistributed after obtaining ex-post values to optimize efficiency.

SDG 13

See ex-antes for full equations.

GS11212

SDG 13: Climate Action	Description		Unit	Source
		2021: 4363 tCO2e		
		2022: 9760 tCO2e		
		2023: 9,999 tCO2e		
		2024: 9,953 tCO2e		
$E_{r,y}$	CO2 emission reductions per year	2025: 9,907 tCO2e	VER	Ex antes

SDG 13: Climate Action	Description		Unit	Source
		2021: 4,194 tCO2e		
		2022: 9760 tCO2e		
		2023: 9,612 tCO2e		
		2024: 9,568 tCO2e		
$E_{r,y}$	CO2 emission reductions per year	2025: 9,523 tCO2e	VER	Ex antes

GS11213

SDG 13: Climate Action	Description		Unit	Source
		2021: 5414 tCO2e		
		2022: 9760 tCO2e		
		2023: 10,000 tCO2e		
		2024: 10,000 tCO2e		
$E_{r,y}$	CO2 emission reductions per year	2025: 9,970 tCO2e	VER	Ex antes

SDG 13: Climate Action	Description		Unit	Source
Er,y	CO2 emission reductions per year	2021: 5,204 tCO2e 2022: 9,760 tCO2e 2023: 9,706 tCO2e 2024: 9,646 tCO2e 2025: 9,584 tCO2e	VER	Ex antes

B.6.4. Summary of ex ante estimates of each SDG outcome

SDG 1: No Poverty

Year	Baseline estimate	Project estimate	Net benefit
Year 1	0% reduction in time spent collecting firewood	1.4 hours reduction in time spent collecting firewood	Reduction in time spent collecting firewood
Year 2	0%	1,4 hours	1,4 hours
Year 2	0%	1,4 hours	1,4 hours
Year 4	0%	1,4 hours	1,4 hours
Year 5	0%	1,4 hours	1,4 hours
Total			

SDG 3: Good Health and Wellbeing

Year	Baseline estimate	Project estimate	Net benefit
Year 1	0% reduction in exposure to Household Air Pollution	Reduction in exposure to Household Air Pollution	Reduction in exposure to Household Air Pollution
Year 2	0%	GS11212 62.4906% GS11213 56.4204%	GS11212 62.4906% GS11213 56.4204%
Year 2	0%	GS11212 62.4906% GS11213 56.4204%	GS11212 62.4906% GS11213 56.4204%
Year 4	0%	GS11212 62.4906% GS11213 56.4204%	GS11212 62.4906% GS11213 56.4204%
Year 5	0%	GS11212 62.4906% GS11213 56.4204%	GS11212 62.4906% GS11213 56.4204%

Total			
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Total number of crediting years	5		
Annual average over the crediting period		GS11212 62. 4906 % GS11213 56. 4204 %	GS11212 62. 4906 % GS11213 56. 4204 %

SDG 7: Affordable and Clean Energy

Year	Baseline estimate	Project estimate	Net benefit
Year 1	0 active ICS	GS11212: 1845 GS11213: 2070	GS11212: 1845 GS11213: 2070
Year 2	0	GS11212: 1845 GS11213:2070	GS11212: 1845 GS11213:2070
Year 3	0	GS11212: 1845 GS11213:2070	GS11212: 1845 GS11213:2070
Year 4	0	GS11212: 1845 GS11213:2070	GS11212: 1845 GS11213:2070
Year 5	0	GS11212: 1845 GS11213:2070	GS11212: 1845 GS11213:2070
Total	0	GS11212: 9,225 stoves GS11213: 10,350 stoves	GS11212: 9,225 stoves GS11213: 10,350 stoves
Total number of crediting years	5		
Annual average over the crediting period	0 active ICS	GS11212: 1845 GS11213:2070	GS11212: 1845 GS11213:2070

SDG 13: Climate Action

Year ⁵	Baseline estimate	Project estimate	Net benefit
Year 1	0	GS11212: 4363 <u>4194</u> tCO2e GS11213: 52044 <u>414</u> tCO2e	GS11212: 4 194 <u>363</u> tCO2e GS11213: 5 2044 <u>14</u> tCO2e
Year 2	0	GS11212: 9,760 tCO2e GS11213: 9,760 tCO2e	GS11212: 9,760 tCO2e GS11213: 9,760 tCO2e
Year 3	0	GS11212: 9, 612 <u>999</u> tCO2e GS11213: 10,097 <u>9706</u> tCO2e (capped at 10,000)	GS11212: 9, 612 <u>999</u> tCO2e GS11213: 10,097 <u>9706</u> tCO2e (capped at 10,000)
Year 4	0	GS11212: 9, 568 <u>953</u> tCO2e GS11213: 10,034 <u>9646</u> tCO2e (capped at 10,000)	GS11212: 9, 953 <u>568</u> tCO2e GS11213: 10,034 <u>9646</u> tCO2e (capped at 10,000)
Year 5	0	GS11212: 9, 523 <u>907</u> tCO2e GS11213: 9, 970 <u>584</u> tCO2e	GS11212: 9, 523 <u>907</u> tCO2e GS11213: 9, 970 <u>584</u> tCO2e
Year 6	0		
Total		GS11212: 42,657 <u>3982</u> GS11213: 45275 (45144 with above 10,000 caps) <u>43,900</u> tCO2	GS11212: 42,657 <u>3982</u> GS11213: 45275 (45144 with above 10,000 caps) <u>43900</u>

Total number of crediting years	5
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Annual average over the crediting period		GS11212: 8531796 GS11213: 87809055	GS11212: 85318796 GS11213: 87809055
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B.7. Monitoring plan

B.7.1. Data and parameters to be monitored

SDG 1

Data / Parameter	Tp,y
Unit	Hours
Description	Project time spent collecting firewood in year y
Source of data	Project Survey
Value(s) applied	Estimated at 1.4 hours. Actual value to be provided in time for each verification.
Measurement methods and procedures	Established through questions in the project survey on a representative sample of the end users.
Monitoring frequency	Annual
QA/QC procedures	Transparent data analysis and reporting
Purpose of data	Calculation on SDG 1 impact. To calculate how much time reduction those who collect firewood experience, which they can reinvest in activities that contribute to their wellbeing.
Additional comment	-

Data / Parameter	Tusage
Unit	NA
Description	Usage by time saved in project activity
Source of data	Project Survey
Value(s) applied	To be provided for each verification
Measurement methods and procedures	Established through questions in the project survey on a representative sample of the end users.
Monitoring frequency	Annual

QA/QC procedures	Transparent data analysis and reporting
Purpose of data	Calculation on SDG 1 impact. To calculate how much time reduction those who collect firewood experience, which they can reinvest in activities that contribute to their wellbeing.
Additional comment	-

SDG 13

Data / Parameter	$N_{p,y}$
Unit	Project Technology Days
Description	Number of cookstoves of each age group operational in year y
Source of data	Cookstove Database
Value(s) applied	To be monitored
Measurement methods and procedures	Sum of the total number of cookstoves distributed multiplied by the number of days crediting each technology earns in a given monitoring period
Monitoring frequency	Continuous recording
QA/QC procedures	Sales records are analysed prior to calculation of parameter
Purpose of data	Calculation of emission reductions
Additional comment	-

Data / Parameter	$U_{p,y}$
Unit	Percentage
Description	Usage rate in project scenario p through year y
Source of data	Annual Usage Survey
Value(s) applied	Estimated at 90%. To be monitored
Measurement methods and procedures	Usage Survey
Monitoring frequency	Annual
QA/QC procedures	Transparent data analysis and reporting
Purpose of data	Calculation of usage
Additional comment	A usage parameter is derived for every age group of project cookstove being credited.

SDG 3

Data / Parameter	HARP,y
Unit	percentage
Description	Total reduction in Household Air Pollution for project activity in year y (%)
Source of data	Usage Survey and baseline kitchen survey
Value(s) applied	Estimated at 62.4906% (GS11212) and 56.4204% (GS11213). To be monitored
Measurement methods and procedures	The average difference in fuel use per technology per day in the project will be subtracted from the baseline figure. This difference will then be used to calculate the % decrease in fuel use, which is the figure used to calculate the VPAs' contribution to the SDG target
Monitoring frequency	Measured annually
QA/QC procedures	
Purpose of data	To calculate the overall reduction in household air pollution from the project activity compared to the baseline scenario
Additional comment	-

SDG 7

Data / Parameter	$N_{p,y}$
Unit	Number of project cookstoves credited (units)
Description	Cookstove in the project database for project scenario p through year y
Source of data	Total Sales Record
Value(s) applied	GS11212: 1845 (estimated) GS11213: 2070 (estimated)
Measurement methods and procedures	The Sales Record of the Project will be updated at least annually.
Monitoring frequency	Continuous
QA/QC procedures	Transparent data analysis and reporting

Purpose of data	To determine number of Active ICS disseminated by project
Additional comment	The total sales record is divided based on project scenario to create the project database

Data / Parameter	DFn
Unit	Fraction
Description	Discount factor to account for efficiency loss of project cookstoves
Source of data	Default value 0.99 i.e, 1% efficiency loss per year
Value(s) applied	
Measurement methods and procedures	
Monitoring frequency	Annual
QA/QC procedures	Transparent data analysis and reporting
Purpose of data	
Additional comment	This default can be used if stoves are found in good condition during annual surveys. For each year, the stoves of the age-group x-y should be physically verified. In the case of progressive installations, stove of agegroup 0-1 shall also be shall be physically verified each year through random sampling approach. Please follow the survey format B to capture the required information. Minimum number of sample size shall be selected following the guidelines provided in section 4.2, option b. During annual surveys if it is found that the project cookstoves are not in working conditions, proportionate population of project cookstoves should be excluded from the project database, until these cookstoves are not replaced with new cookstoves. A site visit by an Objective Observer with relevant technical background would be required at the time of first internal verification and then subsequently after every 2 years from the previous issuance.

Data / Parameter	DFP,stove,y
Unit	Fraction
Description	Discount factor to account for the baseline stove use in project scenario p during the year y
Source of data	Monitoring surveys
Value(s) applied	10%
Measurement methods and procedures	
Monitoring frequency	Annual
QA/QC procedures	Transparent data analysis and reporting
Purpose of data	
Additional comment	The discount factor for baseline-stove use may be determined based on number of meals cooked using baseline stove. The required information shall be captured through sample surveys carried out following random sampling approach for each age-group of the project stove. Minimum number of sample size shall be selected following the guidelines provided in section 4.2, option b. Please refer to the survey format B, Annex A for sample questions to capture this information.

B.7.2. Sampling plan

Cross sampling of devices will be applied across all homogenous VPAs in the project area. Homogenous VPAs are defined as those that are sharing a common baseline. Both Project Stoves share a common baseline and will be monitored separately. Sampling will adhere to the 90/30 precision rules outlined.

Individual participants will be randomly selected from the cookstove electronic database. Sample sizes will be in line with the Gold Standard requirements. The random sample group is reselected for every monitoring period to ensure the selection remains random.

The surveys below will be monitored under the cross-sampling approach:

- Annual Usage Survey/Monitoring survey

The surveys will be conducted so as to ensure that they are within the end date of the respective monitoring periods for each VPA.

B.7.3. Other elements of monitoring plan

This VPA, and any future homogenous VPAs, will be filled with improved stoves sold under the project, each with a unique identification number. Purchasers of the stove will be informed about the carbon transfer

Estufas Chispa (registered under Soluciones Apropriadas S.A) are responsible for the production, distribution and operational management of the stoves. Monitoring might be conducted by Estufas Chispa or outsourced to other professional third parties. They also keep provisions for data archiving, in paper or through mobile monitoring.

All data collection is used solely for the monitoring of the project, and will not be shared with third parties.

Sales Record

During the sales process, representatives of Estufas Chispa will collect data and educate recipients on proper usage of the technology.

A Sales Record will record the following information and will be backed up electronically. The representatives will endeavor to collect all of the following information in order to build a comprehensive Sales Record. Though due to the nature of stove dissemination in this Project, it may not be feasible for the vendor to collect all information. As per the methodology the Sales Record will however ensure representativeness and that "The names and telephone numbers or name and addresses collected must be as many as commensurate with representative scale, i.e. the names and addresses or phone numbers where possible) within sales record shall be large enough so that surveys can be based on representative, randomly selected samples".

The following data will be collected:

- Date of sale
- Geographic area of sale
- Model/type of project technology sold
- Quantity of project technologies sold

- Name, address and telephone number of end user (where feasible)
- Unique stove ID, equal to the owner's national personal ID number

Carbon Transfer Form

The vendee will be informed about the transfer of carbon rights.

The details will be added to the electronic database by Estufas Chispa, which will provide the value to randomly select the monitoring samples.

Signatures of the carbon transfer forms may have signed before or after distribution.

For stoves that have been included retroactively in the project, users have been informed after distribution. Crediting of the stove will only start from the signature date.

Project Database

The Project Database is derived from the Sales Record. The Database tracks the active stoves which provides the value for PTDs and is used to randomly select the monitoring samples.

Ongoing Monitoring Studies

All monitoring studies will be recorded on a hard copy which the monitoring team will transfer into digital format for analysis documentation and reporting by Estufas Chispa. Each of these surveys will be carried out for each project technology.

- Usage/Monitoring Survey
Completed annually.
- Begins within 1 year after project registration.
- Surveys end users currently using project technologies to explore changes in project scenario over time.
- Kitchen observation – usage surveys will be carried out in each household in order to gather objective information.
- Interview with the primary cook – each usage survey interview will be carried out to gather information on stove use patterns
- Photos of the cooking area(s) – a photograph will be taken of the cooking area at the end of each usage survey upon consent of the interviewee
- GPS co-ordinates – GPS co-ordinates will be gathered during each usage survey

SECTION C. DURATION AND CREDITING PERIOD

C.1. Duration of project

C.1.1. Start date of VPA

This is a retroactive project with start date 28/10/2020

C.1.2. Expected operational lifetime of VPA

15 years (5 years twice renewable)

C.2. Crediting period of project

C.2.1. Start date of crediting period

The crediting period begins one day after the project start date: 29/10/2020
(estimated)

C.2.2. Total length of crediting period

5 years twice renewable.

SECTION D. SUMMARY OF SAFEGUARDING PRINCIPLES AND GENDER SENSITIVE ASSESSMENT

D.1. Safeguarding Principles that will be monitored

A completed Safeguarding Principles Assessment is in [Appendix 1](#), ongoing monitoring is summarised below.

Principles	Mitigation Measures added to the Monitoring Plan
Principle 3: Community Health, Safety and Working Conditions	Will be measured through SDG 3
Principle 5: Corruption:	
The project shall not involve, be complicit or inadvertently contribute to or reinforce corruption or corrupt projects.	TBC in the Local Stakeholder Consultation, which is currently postponed due to Covid-19 applying the Interim Measures. We will collect feedback when appropriate on this matter by stakeholders from all categories.

D.2. Assessment that project complies with GS4GG Gender Sensitive requirements

Question 1 - Explain how the project reflects the key issues and requirements of Gender Sensitive design and implementation as outlined in the Gender Policy?	The project will reduce the amount of resources required to cook. There will be less burden on women, men and children, as less firewood for cooking needs to be collected. This will reduce the time burden on women and men in the socially isolating activity of collecting resources. Women are encouraged to attend the LSC, and express their views and concerns of the Project. Women and men will have more time to engage in activities due to the reduced time required to be spent gathering firewood. Both women and men will benefit from the project activities, no group is excluded from participating in the project activities or use of the stoves. The project will decrease the workload of women in collecting firewood and time
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	spent cooking, thereby allowing more time to engage in other activities. The project aims to benefit the whole community equally and women's equal participation in the LSC is encouraged. The Project will increase natural protection as less firewood will be required for cooking. No further risks or hazards for women were identified.
Question 2 - Explain how the project aligns with existing country policies, strategies and best practices	Official government documentation on Health and Wellbeing in Guatemala includes the 'Sustainable Health Agenda for the Americas 2018-2030' as the political strategy that will provide the development in the region for the next decade. The five main areas of focus include establishing developments for a healthier and equal continent, reaffirming the right to health and ensuring equality in social inclusion, working on the continuous efforts for universal health, developing systems that are resilient and are able to address concern, collaboration with the Pan-American Health Office. The document (available at: https://iris.paho.org/bitstream/handle/10665.2/49169/CSP296-spa.pdf?sequence=1&isAllowed=y) lays out the declarations of the ministries and secretariats of the region's countries, including Guatemala. The project will improve the health of its cookstove users. Cooking is nearly always carried out by women and therefore women are the ones who are subjected to indoor air pollution which seriously affects their health. By using our stove we can remove 99% of the smoke from the house and reduce the risk of respiratory related illness burns to women and children. The improved cookstoves will reduce the burden on both women and men of to collect firewood, additionally women in particular benefit from reduce time cooking and exposure to smoke. This gives them more time and opportunity to

	pursue development enhancing activities such as income generating activities and education, which may have once been sacrificed to provide firewood for the homestead. This acts to close the gender gap by allowing women and girls more time for self-improvement and development. The project survey will monitor the impacts of time saved from the project and what activities respondents are choosing to do with their time saved. This helps beneficiaries achieve a more sustainable livelihood over the project lifetime.
Question 3 - Is an Expert required for the Gender Safeguarding Principles & Requirements?	No – an expert is not required as the project developer ensures the project meets any public policy for gender equality and women’s empowerment as per the ‘Gender Safeguarding Principles & Requirements’.
Question 4 - Is an Expert required to assist with Gender issues at the Stakeholder Consultation?	No – the project developer may refer to the Committee on the Elimination of Discrimination against Women to avoid any ‘Gender issues at the Stakeholder Consultation’.

SECTION E. SUMMARY OF LOCAL STAKEHOLDER CONSULTATION

The below is a summary of the 2 step GS4GG Consultation for monitoring purposes. Please refer to the separate Stakeholder Consultation Report for a complete report on the initial consultation and stakeholder feedback round.

E.1. Summary of stakeholder mitigation measures

An in person Local Stakeholder Consultation has not been conducted yet, due to in-country covid restrictions. The meeting has been postponed according to the interim measures, until the situation allows.

A virtual LSC is taking place on 06/10/2022, to present the project and receive feedback.

E.2. Final continuous input / grievance mechanism

Method	Include all details of Chosen Method (s) so that they may be understood and, where relevant, used by readers.
Continuous Input / Grievance Expression Process Book (mandatory)	A Grievance Expression Process Book is held at the office of our local partner Estufas Chispa. This will allow community members without access to a telephone or the internet to send comments to the organisations implementing the project.
GS Contact (mandatory)	help@goldstandard.org
Whatsapp	Stakeholders can get in touch with Estufas Chispa's Customer Service chat, through +50249734331. This is usually the preferred communication channel. This same number can be called and access through payphones in the communities.
Telephone	CO2balanceUK LtdProject Manager: +44 1823 332 233 Stakeholders with telephone access may find this the most convenient way to contact the project partners.
Social Media	https://www.instagram.com/estufaschispa/ https://www.instagram.com/co2_balance/ Stakeholders can provide feedback and/or concerns through instagram

SECTION F. Eligibility and inclusion criteria for VPAs inclusion

The below table shall be completed for all VPAs.

The CME shall provide clear description on how eligibility criteria set at real case VPAs are complied with for each real case and regular VPAs submitted for inclusion.

The CME shall not change the eligibility criteria and required condition set at real case VPAs. At the time of inclusion of regular VPAs, the CME shall only describe how the regular VPAs comply with the eligibility criterion.

No.	Eligibility Criterion	Description/ Required condition	Description of the VPA in relation to the criteria, Means of Verification and Supporting evidence for inclusion
1	Types of Project	Eligible Projects shall include physical action/implementation on the ground. Pre-identified eligible Project types are identified in the Eligibility Principles and Requirements section.	The projects involve the sale of locally designed and manufactured improved cookstoves. The project type is eligible under Community Services Activity Requirements 3.1.1(b) End-Use Energy Efficiency: efficient cooking.
2	Location of Project	Projects may be located in any part of the world.	The VPA is located in Guatemala as specified in Section A.2 of this document. Guatemala is one of the countries outlined in the PoA-DD. The VPA is unlikely defined by a range of GPS coordinates and current administrative maps to define the project boundary.
3	Project Area, Project Boundary and Scale	The Project Area and Project Boundary shall be defined. Projects may be developed at any scale although certain rules, requirements and limitations may apply under specific Activity Requirements, Impact Quantification Methodologies and Products Requirements.	The project boundary, including GPS co-ordinates and maps are included in Section A.2 and B.3. This VPA is not included by any other carbon standard. The host country, Guatemala, and location of each VPA is specified in Section A.2, in line with the locations outlined in Section A.3 of the PoA-

		In order to avoid double counting the Project shall not be included in any other voluntary or compliance standards programme unless approved by Gold Standard (for example through dual certification). Also if the Project Area overlaps with that of another Gold Standard or other voluntary compliance standard programme of a similar nature, the project shall demonstrate that there is no double counting of impacts at design and performance certification (for example use of similar technology or practices through which the potential arises for double counting or misestimation of impacts amongst projects).	DD. The project boundary, including GPS co-ordinates and maps are included in Sections A.2 and B.3.
...	(d) Host Country Requirements	Projects shall be in compliance with applicable Host Country's legal, environmental, ecological and social regulations.	The Project Activity does not violate any of the Host Country's legal, environmental, ecological and social regulations.
	(e) Contact Details	As part of the Project Documentation the Project Developer shall provide (i) name and (ii) contact details of all Project Participants; AND in case of an organisation (iii) the legal registration details and (iv) documentation by the governing jurisdiction that proves that the entity is in good standing (defined as being a legal or other appropriate entity registered in or allowed to operate within the required jurisdiction and with no evidence of insolvency or legal/criminal notices placed against it or any of its Directors). Gold	Contact details of the Project Developer are included in Appendix 1.

	Standard retains the right (at its own discretion) to refuse use of the Standard where reputational concerns are highlighted.	
(f) Legal Ownership	Full and uncontested legal ownership of any Products that are generated under Gold Standard Certification, (for example carbon credits) shall be demonstrated. Where such ownership is transferred from project beneficiaries this must be demonstrated transparently and with full, prior and informed consent (FPIC). Note that for certain Project types there is a requirement for full and uncontested legal land title/tenure to be demonstrated. These are contained within specific Activity or Product Requirements. All projects shall immediately report to Gold Standard any land title/tenure disputes arising.	At the point of sale, purchasers of the stove are informed of the transfer of carbon rights. The terms of the carbon transfer are stated on the CTF.
(g) Other Rights	As well as legal title and ownership, the Project Developer shall also demonstrate where required uncontested legal rights and/or permissions concerning changes in use of other resources required to service the Project (for example, access rights, water rights etc.). Any known disputes or contested rights must be declared immediately to Gold Standard by the Project Developer and resolved prior to further Project	There are no disputes or contested rights that have been identified in relation to rights relevant to the project activity

	implementation in affected areas.
(h) Official Development Assistance (ODA) Declaration	All Project Developers A declaration confirming applying for project that there is no diversion activities located in a of ODA has been country named by the submitted. OECD Development Assistance Committee’s ODA recipient list and seeking Gold Standard Certification for carbon credits shall declare the Official Development Assistance (ODA) support. The Project Developer shall follow the GHG Emissions Reduction & Sequestration Product Requirements and submit the declaration at the time of Design Certification.

APPENDIX 1 - SAFEGUARDING PRINCIPLES ASSESSMENT

Complete the Assessment below and copy all Mitigation Measures for each Principle into [SECTION D](#) above. Please refer to the instructions in the [Guide to Completing](#) this Form below.

Assessment Questions/ Requirements	Justification of Relevance (Yes/potentially/no)	How Project will achieve Requirements through design, management or risk mitigation.	Mitigation Measures added to the Monitoring Plan (if required)
Principle 1. Human Rights			
1. The Project Developer and the Project shall respect internationally proclaimed human rights and shall not be complicit in violence or human rights abuses of any kind as defined in the Universal Declaration of Human Rights	No	The project will adhere to all human rights requirements including respecting internationally proclaimed human rights and Universal Declaration of Human Rights and will not discriminate in any way.	
2. The Project shall not discriminate with regards to participation and inclusion	No	The project shall ensure that everyone in the baseline within the project area has equal opportunity to participate and be included in the project.	

Principle 2. Gender Equality			
1. The Project shall not directly or indirectly lead to/contribute to adverse impacts on gender equality and/or the situation of women	No	Both women and men will benefit from the project activities, no group is excluded from participating in the project activities or use of the stoves.	
2. Projects shall apply the principles of non discrimination, equal treatment, and equal pay for equal work	No	The project decreases the workload of women in collecting firewood and cooking, thereby allowing more time to engage in other activities.	
3. The Project shall refer to the country's national gender strategy or equivalent national commitment to aid in assessing gender risks	No	No further risks or hazards for women and girls have been identified.	
4. (where required) Summary of opinions and recommendations of an Expert Stakeholder(s)	No		
Principle 3. Community Health, Safety and Working Conditions			
1. The Project shall avoid community exposure to increased health risks and shall not adversely affect the health of the	Yes	The project will reduce the community exposure to illnesses associated with household air pollution	

workers and the community		through the distribution of improved cookstoves.	
Principle 4.1 Sites of Cultural and Historical Heritage			
Does the Project Area include sites, structures, or objects with historical, cultural, artistic, traditional or religious values or intangible forms of culture?	No	The project area does not include cultural and historic sites. The focus of the project is distributing improved cookstoves only.	
>>			
Principle 4.2 Forced Eviction and Displacement			
Does the Project require or cause the physical or economic relocation of peoples (temporary or permanent, full or partial)?	No	The focus of the project is distributing improved cookstoves only.	
>>			
Principle 4.3 Land Tenure and Other Rights			
a. Does the Project require any change, or have any uncertainties related to land tenure arrangements and/or access rights, usage rights or land ownership? b. For Projects involving land use tenure, are there any uncertainties with regards to	No	No changes to land tenure arrangements and/or rights are required.	

land tenure, access rights, usage rights or land ownership?			
>>			
Principle 4.4 – Indigenous people			
Are indigenous peoples present in or within the area of influence of the Project and/or is the Project located on land/territory claimed by indigenous peoples?	Yes	Many of the recipients of the stove are indigenous peoples. They are voluntarily buying or receiving the stove.	
>>			
Principle 5. Corruption			
1. The Project shall not involve, be complicit in or inadvertently contribute to or reinforce corruption or corrupt Projects	Yes	The project shall ensure that all forms of corruption are avoided. Project beneficiaries are able to contact the project developer and implementer through the continuous grievance mechanism to report any form of corruption.	
Principle 6.1 Labour Rights			
1. The Project Developer shall ensure that all employment is in compliance with national	Yes	The project adheres to all labour laws and requirements.	

labour occupational health and safety laws and with the principles and standards embodied in the ILO fundamental conventions 2. Workers shall be able to establish and join labour organisations 3. Working agreements with all individual workers shall be documented and implemented and include: a) Working hours (must not exceed 48 hours per week on a regular basis), AND b) Duties and tasks, AND c) Remuneration (must include provision for payment of overtime), AND d) Modalities on health insurance, AND e) Modalities on termination of the contract with provision	Yes Yes		
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for voluntary resignation by employee, AND f) Provision for annual leave of not less than 10 days per year, not including sick and casual leave. 4. No child labour is allowed (Exceptions for children working on their families' property requires an Expert Stakeholder opinion) 5. The Project Developer shall ensure the use of appropriate equipment, training of workers, documentation and reporting of accidents and incidents, and emergency preparedness and response measures	No Yes		
Principle 6.2 Negative Economic Consequences			
1. Does the project cause negative economic consequences during	No	There are no negative economic impacts and	

and after project implementation?		potential risks to local economy.	
>>			
Principle 7.1 Emissions			
Will the Project increase greenhouse gas emissions over the Baseline Scenario?	No	GHG emissions will be reduced through replacing less efficient cooking methods with improved cook stoves.	
>>			
Principle 7.2 Energy Supply			
Will the Project use energy from a local grid or power supply (i.e., not connected to a national or regional grid) or fuel resource (such as wood, biomass) that provides for other local users?	No	The project will reduce consumption of biomass.	
>>			
Principle 8.1 Impact on Natural Water Patterns/Flows			
Will the Project affect the natural or pre-existing pattern of watercourses, ground-water and/or the watershed(s) such as high seasonal flow variability, flooding potential, lack of aquatic connectivity or water scarcity?	No	There will be no significant change in the natural or pre-existing pattern of watercourses, ground-water and/or watersheds(s).	

>>			
Principle 8.2 Erosion and/or Water Body Instability			
a. Could the Project directly or indirectly cause additional erosion and/or water body instability or disrupt the natural pattern of erosion? b. Is the Project's area of influence susceptible to excessive erosion and/or water body instability?	No	The focus of the project is distributing improved cookstoves only	
>>			
Principle 9.1 Landscape Modification and Soil			
Does the Project involve the use of land and soil for production of crops or other products?	No	No crops or other products are produced in the project.	
>>			
Principle 9.2 Vulnerability to Natural Disaster			
Will the Project be susceptible to or lead to increased vulnerability to wind, earthquakes, subsidence, landslides, erosion, flooding,	No	There will be no impact by the project to natural disasters.	

drought or other extreme climatic conditions?			
>>			
Principle 9.3 Genetic Resources			
Could the Project be negatively impacted by or involve genetically modified organisms or GMOs (e.g., contamination, collection and/or harvesting, commercial development, or take place in facilities or farms that include GMOs in their processes and production)?	No	No GMOs will be used in the project.	
>>			
Principle 9.4 Release of pollutants			
Could the Project potentially result in the release of pollutants to the environment?	No	The Project reduces the amount of pollutants released into the air.	
>>			
Principle 9.5 Hazardous and Non-hazardous Waste			
Will the Project involve the manufacture, trade, release, and/ or use of hazardous and non-hazardous chemicals and/or materials?	No	The Project does not deal with hazardous or non-hazardous chemicals and/or materials.	

>>			
Principle 9.6 Pesticides & Fertilisers			
Will the Project involve the application of pesticides and/or fertilisers?	No	No pesticides and/or fertilisers are to be used in the project.	
>>			
Principle 9.7 Harvesting of Forests			
Will the Project involve the harvesting of forests?	No	The project reduces the consumption of firewood, therefore having a positive impact on forest conservation.	
>>			
Principle 9.8 Food			
Does the Project modify the quantity or nutritional quality of food available such as through crop regime alteration or export or economic incentives?	No	The project has no impact on modifying foods or altering crop regimes.	
>>			
Principle 9.9 Animal husbandry			
Will the Project involve animal husbandry?	No	The project does not involve animal husbandry.	
>>			
Principle 9.10 High Conservation Value Areas and Critical Habitats			

Does the Project physically affect or alter largely intact or High Conservation Value (HCV) ecosystems, critical habitats, landscapes, key biodiversity areas or sites identified?	No	The project decreases the consumption of firewood having a positive impact on conserving forest ecosystems.	
>>			
Principle 9.11 Endangered Species			
a. Are there any endangered species identified as potentially being present within the Project boundary (including those that may route through the area)? b. Does the Project potentially impact other areas where endangered species may be present through transboundary affects?	No	There are several endangered species in Guatemala. The project is not envisaged to have any impact on their habitat as it only affects the household cookstoves. The project does not impact other areas where endangered species are present.	

APPENDIX 2- CONTACT INFORMATION OF VPA IMPLEMENTER

Organization name	CO2balance UK Ltd
Registration number with relevant authority	04903159
Street/P.O. Box	Cook Way
Building	1 Discovery House
City	Taunton
State/Region	Somerset
Postcode	TA2 6BJ
Country	United Kingdom
Telephone	+44 (0) 1823 332233
E-mail	matthew.pike@co2balance.com
Website	www.co2balance.com
Contact person	Matthew Pike
Title	Project Manager
Salutation	Mr
Last name	Pike
Middle name	
First name	Matthew
Department	Projects
Mobile	
Direct tel.	+44 (0) 1823 332233
Personal e-mail	matthew.pike@gmail.com

APPENDIX 3-SUMMARY OF APPROVED DESIGN CHANGES

Please refer to [Design Changes Requirements](#) for more information on procedures governing Design Changes

Revision History

Version	Date	Remarks
2.0	4 May 2022	
1.1	7 October 2020	Hyperlinked section summary to enable quick access to key sections Improved clarity on Key Project Information Inclusion criteria table added Gender sensitive requirements added Prior consideration (1 yr rule) and Ongoing Financial Need added Safeguard Principles Assessment as annex and a new section to include applicable safeguards for clarity Improved Clarity on SDG contribution/SDG Impact term used throughout Clarity on Stakeholder Consultation information required Provision of an accompanying Guide to help the user understand detailed rules and requirements
1.0	10 July 2017	Initial adoption